

GUNA VARDHAN KUMAR

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EDUCATION

Vellore Institute of Technology

B.Tech - Computer Science and Engineering - CGPA - 8.66

2022 – 2026

Bhopal, India

PROJECTS

RescueCall Safety Application 🔗 | Python, Machine learning, API's

March 2025

- Developed an AI-powered mobile application that detects human screams in real time and triggers emergency alerts with GPS location to enhance public safety and crime prevention. Designed and deployed a real-time AI model capable of recognizing human distress screams with high accuracy, reducing response times in emergency scenarios.
- Established a full-stack Flask application with Bootstrap 5 frontend, implementing secure authentication, RESTful APIs, and responsive design with less than 200ms response times.
- Architected ML pipeline using Pandas and Scikit-learn for content-based product similarity, enabling real-time recommendations.

AI-Driven Flight Operations Optimizer 🔗 | NLP, Python

June 2025

- Analyzed flight schedules and radar data to optimize takeoff and landing times, helping reduce congestion at busy airports, such as Mumbai and Delhi. Built an AI-powered interface using NLP to provide real-time query responses and scenario-based schedule impact analysis.
- Developed predictive models to identify flights with high cascading delay risk, enabling targeted interventions for improved operational efficiency.
- Technology Stack : Python, Machine Learning (scikit-learn, pandas), Natural Language Processing, Data Visualization, Open-source AI frameworks, Web scraping, Database management (MongoDB, MySQL).

AI-Driven PDF to podcast platform 🔗 | Natural Language processing, LLM

March 2025

- Developed an AI-based system to convert unstructured PDF documents into engaging, multi-speaker podcast audio using NLP, LLMs, and Text-to-Speech technologies.
- Designed ETL workflows and data warehousing structures to store processed text, audio metadata, and performance metrics for efficient querying and analysis.
- Optimized document processing and audio generation pipelines to support large-scale content conversion with high accuracy and minimal processing time. performance with 95.7% accuracy.

INTERNSHIP

Infosys Springboard 🔗

November 2025 – December 2025

Intern

- Developed a Python + Streamlit-based AudioBook Generator that converts text into audio files with 100% local execution, eliminating backend dependencies and reducing deployment complexity.
- Implemented multilingual Text-to-Speech support for 4 languages, achieving real-time audio generation (12 seconds for 1,000+ characters) with consistent output quality.
- Designed an interactive frontend handling text input, narration enhancement, audio playback, and MP3 export, enabling end-to-end processing in a single workflow with 0 API calls.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Tools: VS Code, Git, Jupyter Notebook, Google Colab, Excel

AI and ML: Scikit-learn, TensorFlow, Pandas, NumPy

Technologies: Data Visualisation tools, LLM

EXTRACURRICULAR

Yophoria Innovation Challenge (AI Agent Engineer Track)

Oct 2025

Finalist

- Developed an Autonomous AI Tutor Orchestrator using FastAPI and LangGraph for multi-tool orchestration and context-driven parameter extraction.

CERTIFICATIONS

- Oracle Data science Professional Certification
- IBM AI Engineering (coursera)